



◆ SECTION 1: Electric Charges & Coulomb's Law (Q1–20)

- Two point charges $q_1 = 2\mu\text{C}$ and $q_2 = 3\mu\text{C}$ are placed 0.5 m apart in vacuum. The force between them is:
A) 0.216 N
B) 0.216×10^3 N
C) 0.216×10^{-3} N
D) 2.16 N
- Three point charges of $1\mu\text{C}$ each are placed at the vertices of an equilateral triangle of side 1 m. Net force on one charge:
A) $k\sqrt{3}\mu\text{N}$
B) $k\mu\text{N}$
C) $k\cdot 2\mu\text{N}$
D) Zero
- Two charges q and $-q$ are separated by distance d . If the medium between them is replaced by a dielectric of constant K , the force becomes:
A) F/K
B) $K\cdot F$
C) F^2/K
D) F
- A charge q is placed at the center of a square of side a . Electric field at the corner is:
A) kq/a^2
B) $kq/(a\sqrt{2})^2$
C) Zero
D) $kq/(2a^2)$
- Two charges are separated by 0.1 m. Force is 36 N. If one charge is doubled, distance halved, new force:
A) 288 N
B) 72 N
C) 144 N
D) 36 N
- Charges $q_1 = +2\mu\text{C}$, $q_2 = -3\mu\text{C}$ separated by 1 m. Find net force on q_2 due to q_1 and another $4\mu\text{C}$ charge at same distance.
- Two charges q_1 and q_2 separated by r . If force is F , then after doubling both charges and tripling distance, new force:
A) $4F/9$
B) $9F/4$
C) $2F$
D) $F/2$
- Three charges $q, -q, q$ in line, equal spacing. Net force on middle charge:
A) Zero
B) kq^2/d^2

- C) $2kq^2/d^2$
 D) $-2kq^2/d^2$
9. Two small spheres, each of mass m and charge q , hang from strings of length l . At equilibrium, strings make angle θ . Expression for q :
 A) $\sqrt{(mg l \tan\theta / k)}$
 B) $\sqrt{(mg / k \tan\theta)}$
 C) $k/mg \tan\theta$
 D) mg/k
10. Two charges separated by distance r in vacuum, force is F . Medium of dielectric constant K is inserted, force becomes:
 A) F/K
 B) $K \cdot F$
 C) F^2/K
 D) F
11. Two small spheres carry equal charge, hung from same point. Strings form angle 60° . If mass = 0.01 kg , $g = 9.8 \text{ m/s}^2$, find charge q .
12. Four point charges $+q$ at corners of square of side a . Force on charge at one corner due to others:
 A) $2\sqrt{2} k q^2 / a^2$
 B) $3 k q^2 / a^2$
 C) $k q^2 / a^2$
 D) $\sqrt{2} k q^2 / a^2$
13. Two charges separated by distance d . The electrostatic potential energy of system:
 A) $k q_1 q_2 / d$
 B) $k q_1 q_2 / 2d$
 C) $q_1 q_2 / d$
 D) Zero
14. Point charge q is placed at center of square of side a . Electric potential at corner:
 A) kq/a
 B) $kq/\sqrt{2}a$
 C) Zero
 D) $kq/(2a)$
15. Two identical charges repel with force F at distance d . Work done in doubling distance:
 A) $F \cdot d$
 B) $F \cdot d/2$
 C) $2F \cdot d$
 D) Zero
16. Two charges $+q$ and $-q$ separated by d . Work done in moving $+q$ along perpendicular bisector to distance $2d$:
 A) $k q^2 / 2d$
 B) $k q^2 / d$
 C) 0
 D) $k q^2 / 4d$
17. Two charges q and $-q$ separated by 1 m . Electric field at midpoint:
 A) $k q / 0.25 \text{ m}^2$
 B) Zero
 C) $k q / 0.5^2$
 D) $k q / 1^2$

18. Charge $+Q$ split into two parts q_1, q_2 at distance r . Total energy minimized when:
- A) $q_1 = q_2$
 - B) $q_1 = 0.25 Q$
 - C) $q_1 = Q$
 - D) $q_1 = 0$
19. Two equal charges are at ends of rod of length L . Potential energy:
- A) $k q^2 / L$
 - B) $k q^2 / 2L$
 - C) Zero
 - D) $k q^2 L$
20. A charge q is placed at distance x from another charge Q . Work done in moving q to infinity:
- A) $k Q q / x$
 - B) $k Q q / 2x$
 - C) Zero
 - D) $k Q q / \sqrt{x}$

◆ SECTION 2: Electric Field & Potential (Q21–50)

21. Electric field at distance r from long wire with linear charge density λ :
- A) $\lambda / 2\pi\epsilon_0 r$
 - B) $\lambda / 4\pi\epsilon_0 r^2$
 - C) $2\lambda / 4\pi\epsilon_0 r$
 - D) $\lambda / \epsilon_0 r^2$
22. Electric field at center of square due to charges $+q$ at corners:
- A) Zero
 - B) kq / a^2
 - C) $2kq / a^2$
 - D) $kq / \sqrt{2} a^2$
23. Electric field due to ring of charge at axial point:
- A) $k Q x / (x^2 + R^2)$
 - B) $k Q / (x^2 + R^2)$
 - C) $k Q / x^2$
 - D) $k Q R^2 / (x^2 + R^2)^{3/2}$
24. Electric potential at axial point of ring of charge:
- A) $k Q / \sqrt{(x^2 + R^2)}$
 - B) $k Q / x^2$
 - C) Zero
 - D) $k Q x / (x^2 + R^2)$
25. Electric potential at center of uniformly charged ring:
- A) $k Q / R$
 - B) Zero
 - C) $k Q / 2R$
 - D) $k Q / \sqrt{2} R$
26. Two charges $+q$ and $-q$ separated by distance d . Potential at midpoint:
- A) Zero
 - B) kq/d

- C) $-kq/d$
D) $2 kq / d$
27. Electric field inside uniformly charged sphere of radius R at distance r :
A) $k Q r / R^3$
B) $k Q / r^2$
C) Zero
D) $k Q / R^2$
28. Electric field outside uniformly charged sphere at distance $r > R$:
A) $k Q / r^2$
B) Zero
C) $k Q r / R^3$
D) $2 k Q / r^2$
29. Potential at surface of uniformly charged sphere:
A) $k Q / R$
B) Zero
C) $k Q / 2R$
D) $k Q R^2$
30. Potential inside uniformly charged sphere:
A) $k Q (3R^2 - r^2) / 2R^3$
B) $k Q / r^2$
C) $k Q / R$
D) Zero
31. Work done in moving charge along equipotential:
A) Zero
B) Maximum
C) Minimum
D) Depends on path
32. Electric field at equatorial line of dipole at distance $r \gg d$:
A) $-k p / r^3$
B) Zero
C) $k p / r^2$
D) $k p / r^3$
33. Potential due to dipole at equatorial line:
A) Zero
B) $k p / r^2$
C) $k p / r^3$
D) $k p / d^2$
34. Torque on dipole in field E at angle θ :
A) $p E \sin\theta$
B) $p E \cos\theta$
C) Zero
D) $p E \tan\theta$
35. Potential energy of dipole in field:
A) $-p E \cos\theta$
B) $p E \sin\theta$
C) Zero
D) $p E$

36. Electric field at axial point of dipole:
- A) $2 k p / r^3$
 - B) $k p / r^2$
 - C) $k p / r^3$
 - D) Zero
37. Two charges $+q$ and $-q$ separated by 4 cm. Electric field at midpoint:
- A) $k q / 0.04^2$
 - B) 0
 - C) $2 k q / 0.04^2$
 - D) $k q / 0.02^2$
38. Potential energy of two point charges at separation r :
- A) $k q_1 q_2 / r$
 - B) $k q_1 q_2 r$
 - C) $k q_1 q_2 / r^2$
 - D) Zero
39. Electric field due to infinite sheet with surface density σ :
- A) $\sigma / 2\epsilon_0$
 - B) σ / ϵ_0
 - C) $2\sigma / \epsilon_0$
 - D) Zero
40. Electric field between parallel plates with potential difference V , separation d :
- A) V/d
 - B) $V \cdot d$
 - C) d/V
 - D) $2V/d$
41. Equipotential surfaces are:
- A) Perpendicular to field lines
 - B) Parallel to field lines
 - C) Curved only
 - D) Not defined
42. Electric potential at distance r from point charge Q :
- A) $k Q / r$
 - B) $k Q / r^2$
 - C) $k Q r^2$
 - D) Zero
43. Electric field at distance r from line charge:
- A) $\lambda / 2\pi\epsilon_0 r$
 - B) $\lambda / 4\pi\epsilon_0 r^2$
 - C) $\lambda r / 2\pi\epsilon_0$
 - D) Zero
44. Potential at distance r from line charge λ , length infinite:
- A) Undefined (logarithmic)
 - B) $k \lambda / r$
 - C) $k \lambda / r^2$
 - D) Zero
45. A dipole of moment p placed along uniform field. Torque:
- A) Zero
 - B) Maximum

- C) pE
 - D) $-pE$
46. A positive charge moves along electric field line:
- A) Potential decreases
 - B) Potential increases
 - C) Potential constant
 - D) Cannot determine
47. Work done in moving q from potential V_1 to V_2 :
- A) $q(V_2 - V_1)$
 - B) $q(V_1 - V_2)$
 - C) V_1/q
 - D) Zero
48. Electric field inside hollow spherical shell:
- A) Zero
 - B) kQ/r^2
 - C) kQ/r^3
 - D) Maximum
49. Potential inside hollow shell:
- A) Constant
 - B) Zero
 - C) Varies with r
 - D) Maximum at center
50. Electric field at a point due to two equal charges at distance d apart along line joining them:
- A) Vector sum
 - B) Algebraic sum
 - C) Always zero
 - D) Depends on medium